Sarvam AI hackathon · 2026

Bolo Khata

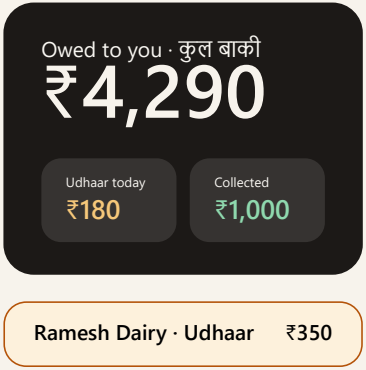
Speak it. The khata writes itself.

बोलो, और खाता लिख गया।

A voice-first credit ledger for India's small shops, built on Sarvam's Indic speech and language models.

Live `bolo-khata-pi.vercel.app`

Code `github.com/DevVaradPatil/bolo_khata_saravam_ai`



The khata is still on paper

- Neighbourhood shops run on **udhaar**: regular customers take goods now and pay later.
- That credit lives in a paper notebook. Entries get missed, totals drift, and pages get lost.
- Ledger apps exist, but they expect **typing in English** with one hand, at a busy counter.
- Owners think and talk in Hindi, Marathi, Hinglish… and in spoken numbers like ढाई सौ.

What the owner actually says

"रमेश डेयरी वाले को साढ़े तीन सौ का उधार लिखो, दूध और पनीर।"

What today's apps need

Open app → find customer → tap credit → type 350 → type note → save

The solution

Speak. Confirm. Done.

1. Bolo

Tap the mic and say it the way you'd say it to a person.

2. Check

The app reads it back and shows a confirm card.

3. Haan

One tap saves it. Totals and history update.

"सुनीता ताईंनी हजार रुपये दिले"

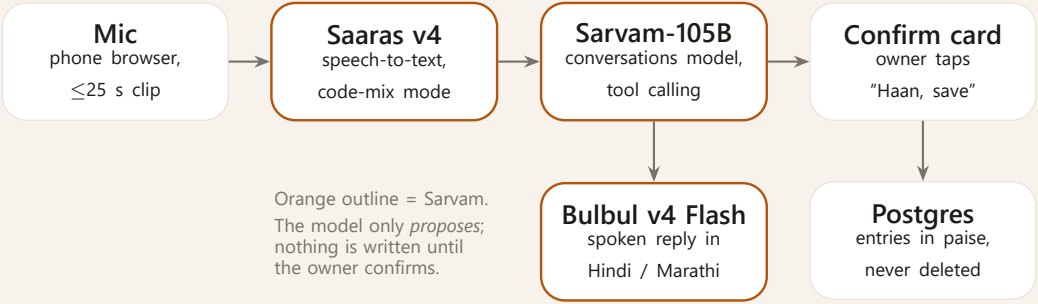
सुनीता पाटिल यांच्याकडून ₹1,000 जमा लिहू का?

Marathi in

spoken reply out

Also: "Priya ka kitna baaki?" · "Aaj kitna udhaar gaya?" · "Naya customer jodo…" · "Last entry cancel karo"

Every AI step is a Sarvam model



An agent that knows when to ask

Tools the model can call

`propose_entry`	credit / payment
`propose_customer`	new customer
`propose_undo`	cancel last entry
`get_balance`	one customer
`list_dues`	who owes most
`get_summary`	today's totals

Several transactions in one sentence become several cards.

Two customers called Ramesh

"रमेश को 200 का उधार लिखो"

रमेश कुमार या रमेश डेयरी, कौन से?

No guess, no card, nothing saved.

Prompt wording alone failed both ways (guessed, or asked "which Ganesh?"). Fix: the **code** works out shared first names from the database and hands the list to the model.

We measured before we built

15 / 15

tool calls correct,
Hindi · Marathi · Hinglish

0.25 s

per agent reply with
sarvam-105b-conversations

₹0.35

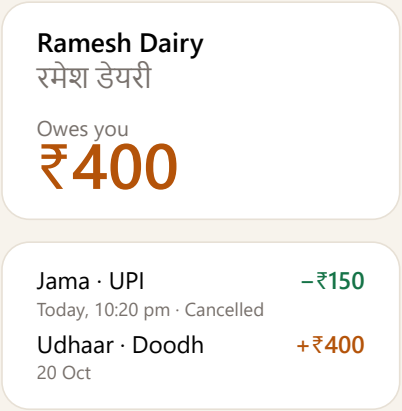
est. cost of one voice turn
(STT + LLM + TTS)

Chat model	Correct	Avg. reply	Note
`sarvam-105b`	14 / 15	2.5 s	once wrote a call as text
`sarvam-105b-conversations`	**15 / 15**	**0.25 s**	chosen

Also found: Saaras keyterm hints turned “Ganesh Kale” into “Ramesh Kale”, so the agent matches names instead.

Built to be trusted with money

- **Confirm before save.** Every change is a card and a spoken read-back.
- **Nothing is deleted.** Undo crosses an entry out; history stays complete.
- **Exact money.** Amounts are stored as whole paise, never decimals.
- **Server re-checks everything** the browser sends back.
- **Bilingual by design.** Names in both scripts; search “Ramesh” or “सुनीता”.
- **Typing fallback** for a noisy counter.



Shipped, live, and small

Stack

Next.js 16 (App Router, streaming) on Vercel
Neon Postgres via Vercel Marketplace
Drizzle ORM · `pg` connection pool
Sarvam JS SDK `sarvamai`

Sarvam APIs in use

`/speech-to-text` · Saaras v4, codemix
`/v1/chat/completions` · tools
`/text-to-speech` · Bulbul v4 Flash
Next: Mayura translate, Sarvam Vision

One voice turn today: **2–5 s** end to end. The biggest pieces are the spoken reply (Bulbul) and the database sitting in the US while users and functions are in India; both have straightforward fixes on the roadmap.

What's next

Reminders

Payment reminder written by **Mayura** in the *customer's* language, plus a **Bulbul** voice note, sent on WhatsApp.

Scan the old khata

Photograph a paper page. **Sarvam Vision** extracts name, amount and date rows to review and import.

Faster & multi-shop

Show the transcript first and stream the voice reply; move the database to India; PIN login per shop; "Aaj ka hisaab" daily summary.

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Try it bolo-khata-pi.vercel.app

Code github.com/DevVaradPatil/bolo_khata_saravam_ai

Powered by Sarvam AI · Saaras · Sarvam-105B · Bulbul